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On the Magnetization History of the Northern Irish Basalts

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Summary

Thermal and alternating field demagnetization studies of 23 basaltic lavas of the Lower Tertiary igneous province of Northern Ireland (the Antrim Basalts) have revealed a very stable but composite magnetization: the resultant appears to consist of a dominating reversed component, thought to be the original one, plus a certain superimposed magnetization of normal polarity. The polarity together with various information about the opaque mineralogy suggest that the latter magnetization, which is the most stable one, has resulted from post-cooling (low temperature) oxidation. A separate component (normal or reversed) appears impossible to estimate with sufficient degree of accuracy but based on a comparison with some previous data it is tentatively suggested that the mean direction of the very stable net magnetization deviates by approximately 20 degrees of arc from that of the original field. On this assumption the magnetization directions derived from the Antrim Basalts would be of doubtful value for instance with regard to the study of any Tertiary continental drift in the North Atlantic region. The remagnetization hypothesis laid down has to be further tested by studying the magnetization of the laterite horizons within the formation which carry a significant amount of haematite formed within the period of lava eruption.

Introduction

The basalts of North-east Ireland, covering an area of about 1500 square miles, represent the southernmost part of the Thulean or North Atlantic igneous province. Outlines of the geology have been given by Charlesworth (1953), Walker (1960) and Wilson & Robbie (1966). The Northern Irish lava succession which was extruded subaerially rests in general upon a thin layer of red clay which in turn overlies the eroded surface of the Cretaceous Upper Chalk. The lava sequence is divided into three subsections: the Lower-, the Tholeiitic- and the Upper Basalts. The Lower- and Upper Basalts are olivine basalts with an average olivine content of 15–20 per cent. Prominent lateritic/bauxitic horizons separate the three series from each other, representing interruptions in the volcanic activity of considerable duration.

The thinner units of the Lower Basalts have a slaggy and visicular appearance and are often badly altered due to zeolitization and weathering. The Tholeiitic Basalts are fine grained and compact and are distinguished by the virtual absence of olivine and by the presence of glass. These latter basalts are thicker than those of the Lower Basalts and may show extensive columnar jointing.

Radiometric age determinations (Miller & Brown 1963) have arrived at an age of 74 My which suggest that the volcanic activity took place at about the Cretaceous/

Tertiary boundary. The estimated age of the Northern Irish basalts is similar to those of Western Scotland, the Faeroes and Eastern Greenland and therefore a comparison of the directions of fossil magnetization between these areas is likely to provide information about the evolution history of the Norwegian Sea since early Tertiary times. Tarling (1967, 1970) has recently attempted such a comparison, but based on the existing data it appears difficult to draw a final conclusion.

In previous palaeomagnetic papers of the Northern Irish basalts (Hospers & Charlesworth 1954; Wilson 1959, 1961a, b, 1962, 1963, 1970) a number of different aspects (palaeomagnetic and geological) have been dealt with. Many of these papers are based on detailed experimental work, but when all available data are seen in conjunction the previous studies give some hints on inherent chemical remagnetization problems, probably resulting in a certain deflection of the original magnetization directions of the lavas. In order to look into this problem a palaeomagnetic rock collection comprising 99 hand samples from 23 different sites (lavas) were taken along the coast from Island Magee in the south to Coleraine in the north. 18 sites (70 samples) originate from the Lower Basalts and the remaining five sites (29 samples) from the Tholeiitic series. On average the flows show tilts of 5–10 degrees and consequently strike and dip of the beds were measured when possible. For orientation of the samples both magnetic compass and sun compass were used.

The magnetic minerals

Ore microscopy

Polished sections from 12 sites have been studied under the ore microscope. In order to distinguish between strongly magnetic and weakly magnetic (or non-magnetic) phases and to detect magnetic inhomogeneities within the grains the magnetic colloid technique have frequently been used. The results are shortly summarized as follows:

(*a*) *The Lower Basalts.* Ilmenite is the dominating ore mineral representing between 50 and 90 volume per cent of the ore mineral content. The ilmenite grains often have intergrowths of wavy haematite blebs. This is probably a result of high temperature oxidation. In some samples the ilmenite grains are rimmed (coated) by a film of probably haematite, sometimes intruding into the grain itself. The olivine grains have been subjected to serpentinization. This process appears to have resulted in the formation of a mixture of iddingsite, goethite, magnetite, haematite and probably maghemite (cation deficient titanomagnetites). The bright bands surrounding the olivine grains attract the magnetic colloid in a few places indicating the presence of magnetite and/or maghemite. The more abundant haematite, which does not attract the magnetic colloid, is optically identified by its colour and anisotropy.

The remaining 10–50 per cent of the ore grains are titanomagnetites, most of which have large exsolved ilmenite lammelae. The total area of these ore minerals attracting the colloid is less than 50 per cent. Some small grains ($< 2\mu$) which were strongly magnetic could not be properly identified.

(b) *The Tholeiitic Basalts.* The ore minerals of these samples seem to consist of more than 98 volume per cent of small ($< 20\mu$) optically homogeneous titanomagnetite grains. Under high magnification (1900×, oil immersion) faint traces of exsolved ilmenite (?) lamellae appear to exist in some grains. All the grains attract the magnetic colloid, but no magnetic patterns have been observed. Less than 2 per cent of the ore minerals is free ilmenite. This latter mineral shows no visible alterations.

Saturation magnetization versus temperature

The saturation magnetization, J_s, as a function of temperature has been measured with a translation balance. The heatings were carried out in air in applied field strengths varying between 3000 and 6500 Oe. The results are as follows:

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(a) *The Lower Basalts.* All samples from this group have Curie temperatures, T_c, in the interval 540–640 °C. The heating and cooling curves are always irreversible. In some cases there are no well-defined T_c as the saturation magnetization is decreasing linearly with temperature. Samples with T_c in the interval 600–640 °C, the most common T_c-range in the Lower Basalts, show a decrease of J_s during cooling together with a systematic lowering of the T_c by about 50–100 °C. A typical example is shown in Fig. 1(a). Curie temperatures in the range of 600–640 °C may be associated with haemo-ilmenites, but owing to the particular change of mineral properties recorded after heating it appears more likely that one is not dealing with true Curie temperatures, but rather with transformation temperatures of cation deficient titanomagnetites (titanomaghemites). The existence of maghemites of high thermal stability has also been reported by Wilson (1961a). In a few samples where the T_c is around 540 °C there are negligible change of this figure even after heating to 700 °C. The cooling curves of these latter samples lie above the heating curves suggesting exsolution-oxidation as the most important ore mineral alteration during heating. The optical observations of haematite are not confirmed by the $J_s - T$ measurements. The reason

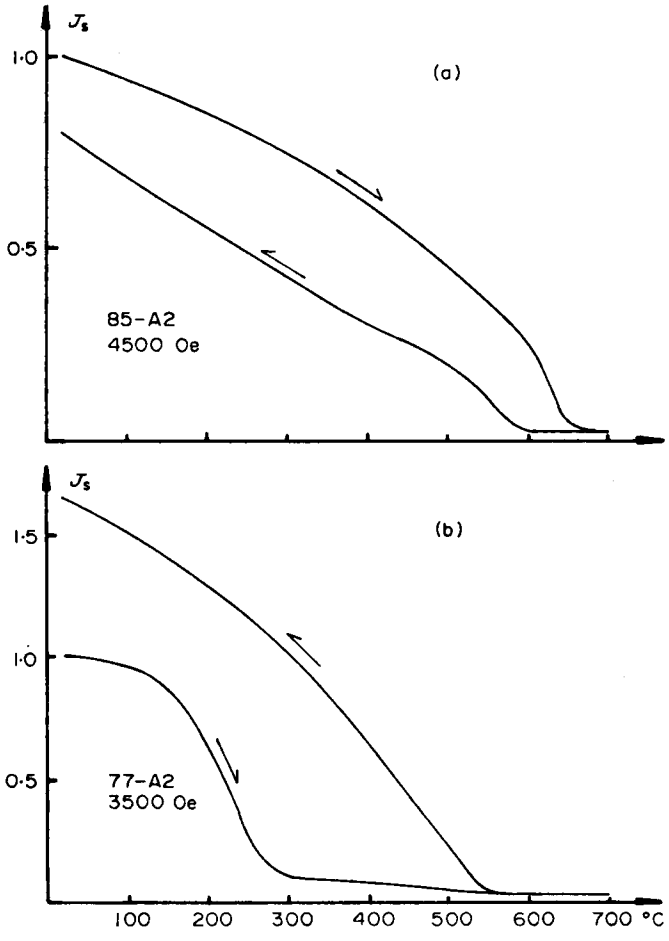


FIG. 1. Examples of saturation magnetization versus temperatures. (a) is a Lower Basalt specimen while (b) is a Tholeiite specimen.

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for this is certainly the relatively small amount of haematite present together with the low specific magnetization of this mineral.

(b) *The Tholeiites.* All samples of this group have single Curie temperatures in the region 220–380 °C. According to Carmichael & Nicholls (1967) this T_c-range corresponds roughly to the composition of the virgin state titanomagnetites in basaltic rocks. A typical example is shown in Fig. 1(b). After heating to 700 °C the T_c increase to around 540 °C while the J_s increases by 20–70 per cent (measured at room temperature). These changes of magnetic properties are also obtained after isothermal heat treatment for a few hours at around 300 °C. The latter rapid J_s increase at a rather low temperature is not compatible with exsolution-oxidation processes in titanomagnetites but rather with unmixing processes in cation-deficient phases.

X-ray diffraction

The X-ray diffraction measurements were done on samples from 10 sites. The samples were crushed with xylene in an agate morter to give grain sizes of approximately 10μ. The diffraction curves obtained were rather complex. However, all samples show peaks of a cubic phase with $R = 8{\cdot}39 - 8{\cdot}50$ Å, suggesting members of the solid solution series magnetite-ulvospinel. In some samples the phase representing the cubic mineral were broad and complex and may indicate the presence of 'maghemites'. Samples belonging to the Lower Basalts show peaks of two rhombohedral phases, identified as ilmenite and haematite. Thus, the X-ray diffraction results are in broad agreement with the ore microscopy and $J_s - T$ measurements.

Ore mineralogy in terms of remanent magnetization

Ore phases from low temperature alterations tend to play an important role in the considered rocks. It seems likely therefore that one or more CRM components are superimposed on the original TRM of these rocks. Whether these chemical magnetizations have upsetting effects on the direction of TRM or not depend critically upon the behaviour of the geomagnetic field during the time of chemical changes (Storetvedt 1970). Thus, if low temperature chemical magnetizations originate in the same dipole field as the TRM no complications in the resultant remanence directions may be generally expected. On the other hand, if relative polar wandering or a change of polarity takes place at the time of CRM acquisition, deviating (false) remanence directions of high magnetic stability may develop. The tholeiites appear to be much less affected by low temperature chemical changes than the Lower Basalts though the X-ray diffraction patterns give some hints on the presence of maghemite. However, the importance of any CRM component in the Northern Irish basalts can only be provided by a careful demagnetization study of the natural remanence.

Demagnetization experiments

A large number of specimens have been subjected to progressive AF and thermal demagnetization. Two different AF equipments and two different furnaces have been used in these studies. The measurements of the remanent magnetization have been carried out by three different observers using two different types of astatic magnetometers. The NRM intensity varies between the orders of 10^{-2} emu cm^{-3} and 10^{-4} emu cm^{-3}, the greatest variation being exhibited by the Tholeiites.

Alternating field results

With some exceptions the Lower Basalts are more resistant against alternating field demagnetization than the Tholeiites. Recalling the difference in state of oxidation

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of these rocks the observed difference in magnetic hardness is as expected (Larson *et al.* 1969). Low stability viscous magnetization plays an unimportant role in the Northern Irish basalts. In practically all specimens studied the characteristic fossil magnetization is directed southerly with a fairly steep upward (negative) inclination. In many specimens these directions remain unchanged even when subjected to demagnetizing fields in excess of 1000 Oe. However, in some specimens more or less systematic movements of the magnetic vector initiate at higher fields. These directional changes are either directed towards north or towards south. Characteristic examples

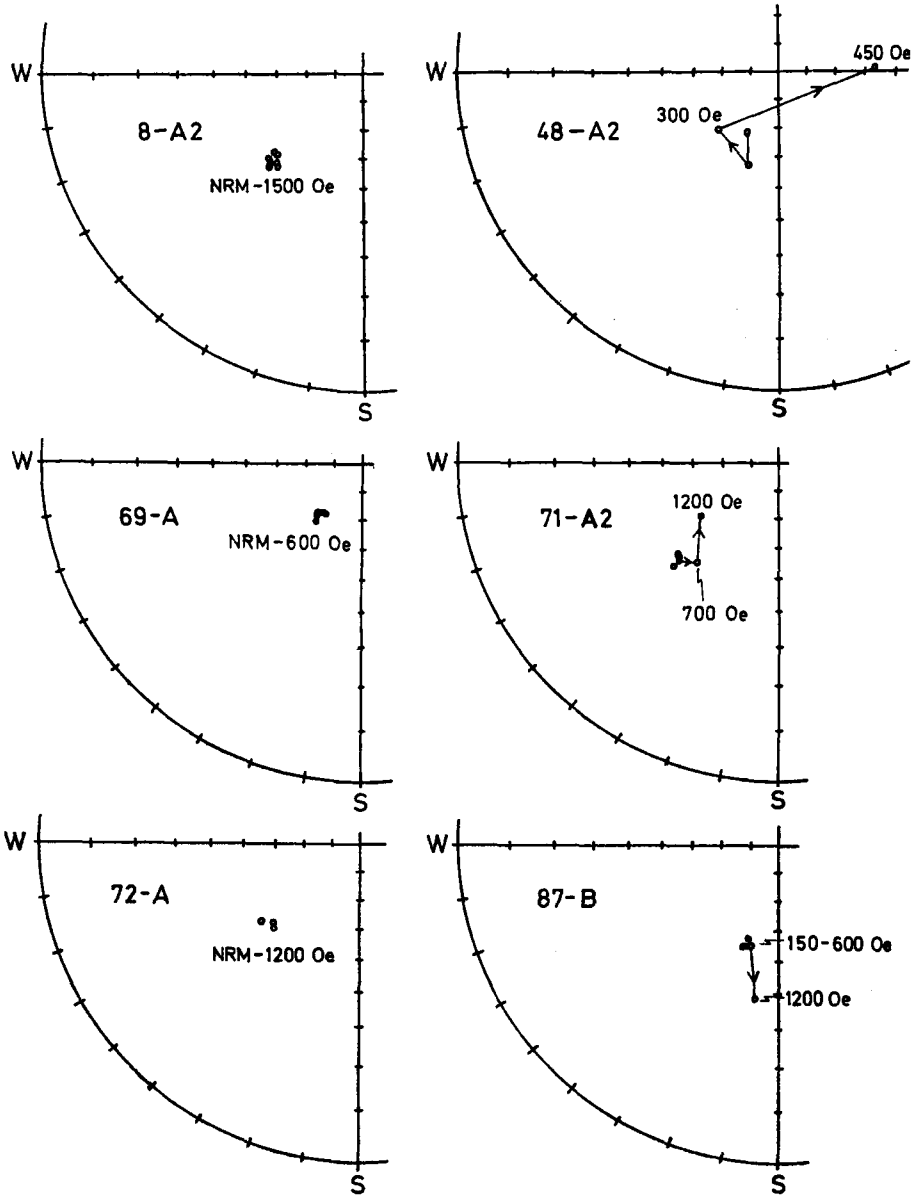


FIG. 2. Characteristic examples of the directional behaviour of the remanent magnetization as a function of increasing alternating fields. All directions are upward seeking.

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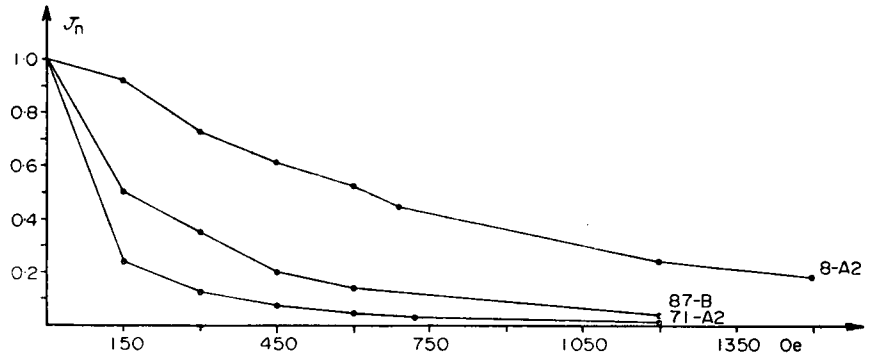


FIG. 3. Examples of remanence intensity versus increasing alternating field.

of directional and intensity behaviour during progressive AF demagnetization are shown in Figs 2 and 3.

Thermal results

The magnetic stability difference between the Lower Basalts and the Tholeiites is very well demonstrated on thermal demagnetization. Thus, the Lower Basalts have an important blocking temperature range between approximately 450 and 650 °C while in the Tholeiites more than 90 per cent of the remanence intensity is in general associated with temperatures below approximately 300 °C (indicating a virgin state titanomagnetite and/or ' maghemite '). The indication of a composite magnetization, as suggested by the directional changes of some specimens during progressive AF demagnetization, has been strongly verified by the thermal demagnetization studies. Thus, in nearly all the Tholeiite specimens tested the remanence vector perform direction changes when demagnetized to temperatures of 300 °C or higher. Similar directional movements are also observed in the Lower Basalts when these are demagnetized to temperatures of 600 °C or higher. The experimental problems of estimating clean high temperature directions are exactly as encountered in previous studies of composite magnetizations (Storetvedt, Halvorsen & Gjellestad 1968, Storetvedt & Halvorsen 1968; Storetvedt & Petersen 1970, 1972; Storetvedt 1970), i.e. the direction in question has to be sought at temperatures near to that at which the total remanence disappears. However, at the temperatures concerned the magnetization is relatively weak and well defined directional trends associated with smooth variation in the corresponding remanence intensity with temperature may be superseded by scattered directional results as well as irregular intensity variation. In the Lower Basalts the critical intensity level here concerned appears to be a factor 10 lower than that of the Tholeiites. This difference is most likely due to the difference in state of oxidation, i.e. the relatively unoxidized (and thermally unstable) Tholeiites being more capable of receiving stray moments caused by small uncompensated magnetic fields acting during the heating experiments. Examples from the demagnetization results are shown by Figs 4 and 5.

From the direction changes on demagnetization (cf. Figs 2 and 4) there appears to be strong indications of the existence of a normally directed high stability component with positive inclination.

The distribution of stable bulk magnetization is given by Fig. 6. These latter directions are based on specimen means in which directional movements after treatment in higher alternating fields or at higher temperatures (low intensity) have been ignored. The bulk directions, irrespective of the level of observation, show a certain SW–NE trend. Tilt correction has no detectable influence on this distribution (including mean direction).

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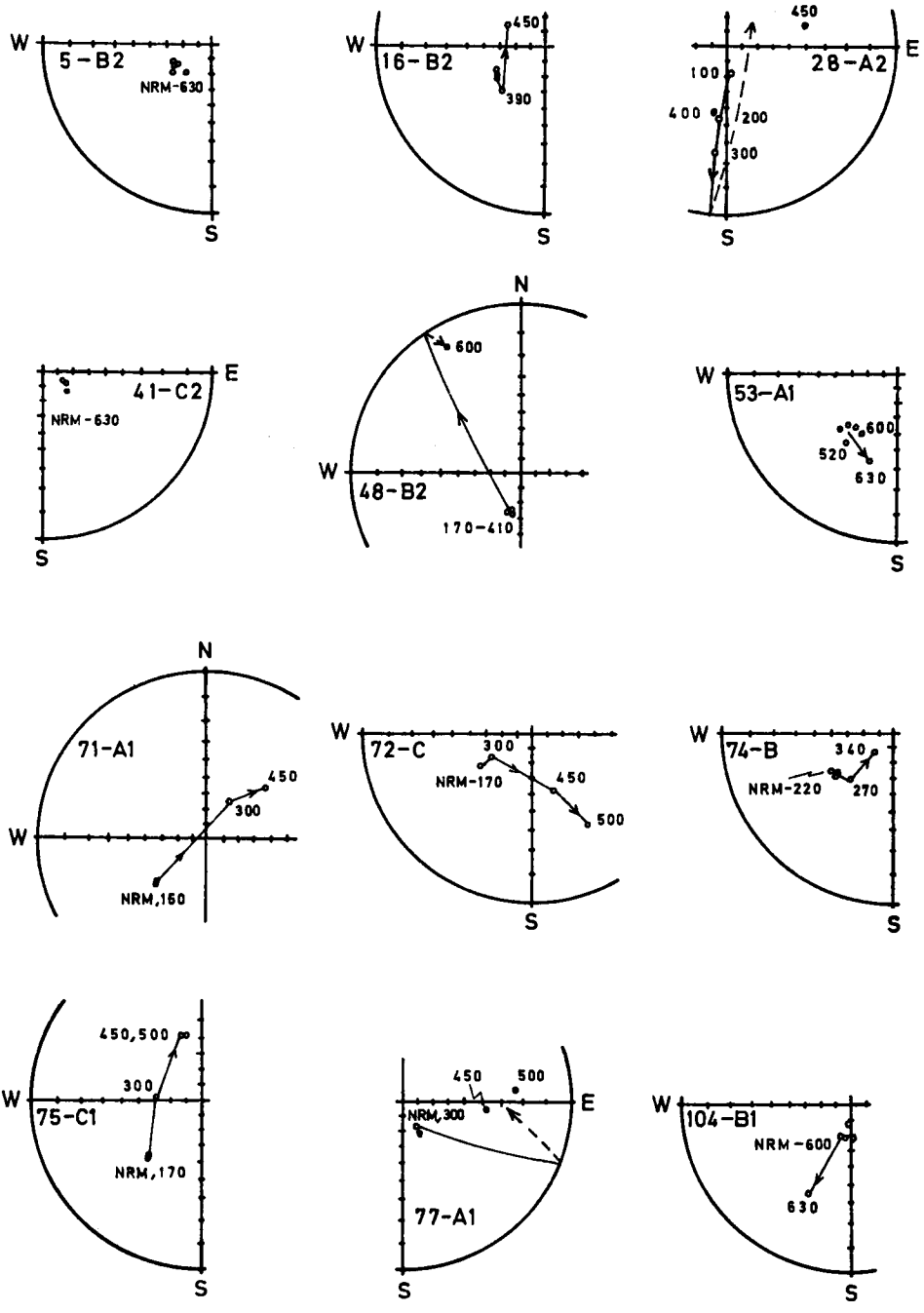


FIG. 4. Examples of remanence directions as a function of temperature. Full circles represent downward inclinations and open circles upward inclination.

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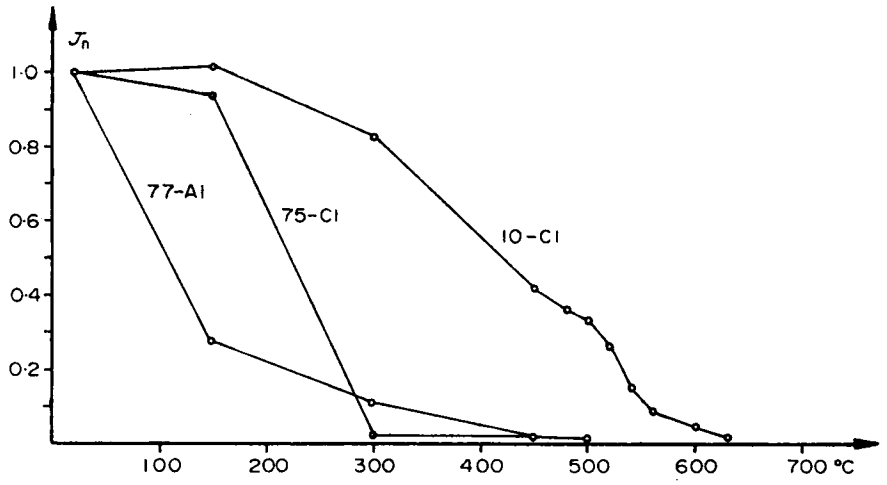


FIG. 5. Representative examples of remanence intensity versus temperature.

Interpretation of experimental data

The demagnetization results together with the available ore mineral data suggest that low-temperature chemical magnetization (CRM) is of significance in the Northern Irish lavas. The resultant magnetic vector seems to be composed of a dominating reversed magnetization plus a certain normally directed component, the latter one representing the most resistant fraction of the fossil magnetization. It appears plausible to suppose that the dominating reversed magnetization was imposed by the original field (i.e. constituting the primary TRM together with any possible CRM subsequently acquired in the same field) while the normal component is regarded to be entirely of secondary origin. It is assumed therefore that the time of chemical changes continued at least into the subsequent normal polarity epoch. A crucial question is whether the net magnetization recorded represents a false (deviating) palaeomagnetic direction or not. Though the present results are in good agreement with those previously obtained (cf. Table 1) the trend of the directions of stable bulk magnetization as here observed (Fig. 6) may throw some doubt about the palaeomagnetic reliability of the Northern Irish lavas. Because of the general experimental problems of estimating single remanence components in partially oxidized rocks an answer to this important question can hardly be provided by the collected material. The most reliable information about the original field direction should emerge either in rocks of which the ore minerals were in a very high state of oxidation prior to

Table 1

Summary of previous palaeomagnetic results from Northern Ireland in comparison with that of the stable bulk magnetization of the present study.

No. of igneous bodies	D	J	α_{95}	Standard deviation	R	Authors
24	194°	−60·2°	5·9°	14°	23·253	Hospers & Charlesworth 1954
41	183°	−64·1°	6·3°	20°	38·45	Wilson 1959
25	184·7°	−54·3°	5·1	13·8°	24·278	Wilson 1970
19	198·5	−59·7°	10·3°	24°	17·44	This paper

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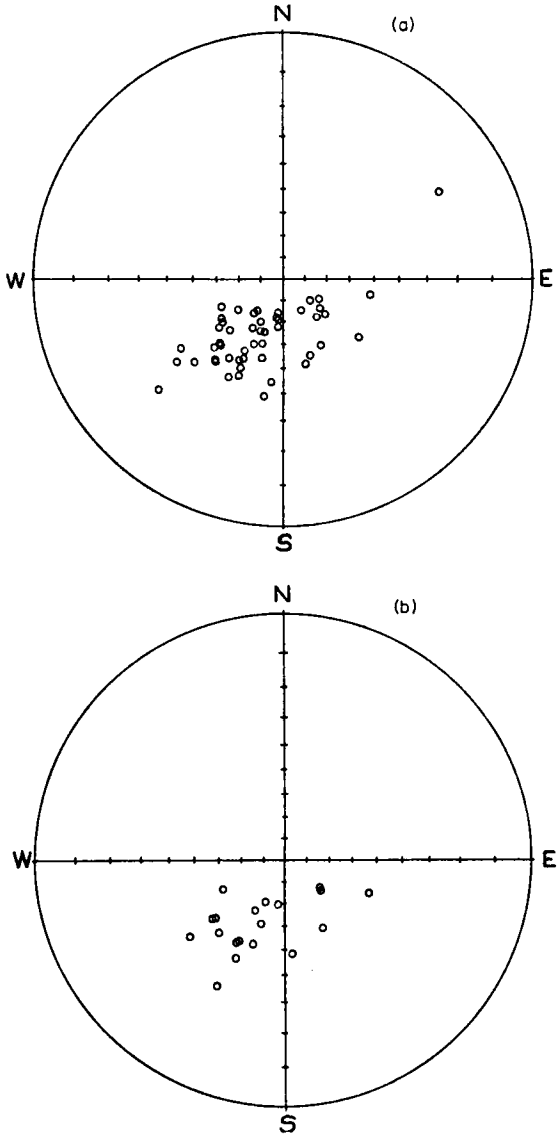


FIG. 6. Directions of stable bulk magnetization representing specimen means (a) and lava means (b) respectively.

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baking or in rocks having undergone severe oxidation within the time covered by the extrusive activity. Fortunately, a few results from such rocks (based on extensive thermal demagnetization) can be picked out from the literature. The palaeomagnetic significance of these and other relevant data are dealt with in the following chapter.

Palaeomagnetic evaluation of some previous data

The first palaeomagnetic data to be considered come from the Interbasaltic Layer exposed at Giant's Causeway in County Antrim. This layer constitutes a sequence of three or four strongly altered lava flows. The outpouring of each of these lavas was followed by a prolonged quiescence in which an extensive weathering took place. According to Eyles (1952) the weathering of each flow appears to have practically reached completion before the onset of the succeeding lava. The process of lateritization has predominantly resulted in bauxite (aluminous laterite) in the interior of the flows changing into a more ferruginous laterite upwards towards the surfaces. Haematite and a fairly stable maghemite seem to be the essential remanence carrying minerals (Wilson 1961a). For reasons not yet properly understood maghemite often give scattered remanence directions, while haematite, which is the stable magnetic end product of oxidation is thought to provide a reliable palaeomagnetic record. One of the laterite horizons of the Interbasaltic Layer appears to be strongly dominated by haematite and should therefore be of vital importance in the present discussion. Thermal experiments (Wilson 1961a, b) suggest that the original chemical remanence of this laterite was completely erased during baking by the overlying flow (the uppermost lava of the Interbasaltic Layer) and replaced by a TRM. Also, the high temperature directions (eight specimens) from this laterite zone are very well grouped and are likely to represent a spot reading of the ancient field at the time of baking. Furthermore, studies of three bauxite specimens from the interior of the baking lava give magnetization directions in excellent agreement with those of the underlying baked laterite. Material from the interior of the lava is not likely to have experienced a significant rise in temperature during formation of the overlying flow (Wilson 1961b). Therefore, the high temperature directions of the bauxite specimens are considered to be of chemical origin and probably formed over a time interval considerably longer than the period of geomagnetic secular variation. This implies that in the bauxite specimens the secular changes of the palaeomagnetic field are expected to be averaged out leaving an axial dipole field reading of the ancient field. For this reason it appears not unlikely that the mean magnetization direction of the considered laterite and bauxite specimens, 178, −41 (with a standard deviation of 8°) represents a good approximation to the dipole field direction in Northern Ireland in early Tertiary times. Directional data from other baked contacts of Northern Ireland are available, but these are not considered here for two reasons: 1. In some cases the ferric iron oxides of the baked rocks have undergone significant reduction processes during heating. Therefore, a subsequent oxidation is likely to have followed whereby the palaeomagnetic reliability of these baked contacts would be as for the basalt lavas dealt with in this paper. 2. Rocks baked by dike intrusions may not give adequate information in the present context as such injections may be considerable younger than the main igneous event.

Thermal demagnetization of some rhyolite specimens (Wilson 1963) tends to support the composite magnetization build-up as shown by the basaltic lavas. Of the four specimens tested three have a reasonable magnetic stability. The directional data of the more stable remanence fraction in two of these latter specimens indicate a gradual change towards a northerly position: when plotted in a north–south, vertical plane the high temperature directions perform a change from intermediate to vertical dips (upward). Microscopic evidence suggests that magnetite and haematite are present in roughly equal proportions (Wilson 1963). It appears pertinent to ask therefore

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whether the major haematite fraction of these rhyolites was formed during some subsequent time while the Earth's magnetic field maintained a normal polarity.

In summary, some previous palaeomagnetic evidence indicates that the dipole field direction for Northern Ireland in early Tertiary times may have been around $D = 180°$, $I = -40°$. Furthermore, there is some evidence of chemical remagnetization having been operating at a later time when the geomagnetic field had changed its sense to a normal polarity.

Conclusions

With reference to the foregoing chapter there are reasons for believing that a direction of about 180, −40 is at present the best estimate of the early Tertiary dipole field for Northern Ireland. The palaeomagnetic reliability of this figure is based on the assumption that the relevant rocks have been in a very high state of oxidation at the end of the original polarity epoch (reversed), making further chemical magnetization difficult to impress. On the other hand, most lavas are far from completely oxidized and therefore they have been capably of acquiring secondary CRMs in subsequent fields. Also, the simplest interpretation of the demagnetization results here presented is that remagnetization in later fields has taken place. Although the secondary CRM contribution is small compared with the 'original TRM/CRM component' the net magnetization appears to be deflected by approximately 20 degrees of arc from the original direction. This deflection seems to be caused by superimposition of a normally directed secondary component though partial remagnetization along more steeply inclined reversed fields can not be entirely ruled out.

The palaeomagnetic pole based on the Tertiary field as here proposed is around 60° N, 175° E, fitting well into the northern extension of the European Palaeozoic polar path, while the pole deduced from the stable bulk magnetization of the present lavas is displaced to a position at approximately 71° N, 126° E (corresponding to an angular distance of about 20 degrees). Whatever the final result will be it appears quite clear that a careful elimination of even relatively small secondary CRMs is of crucial importance for a successful application of Tertiary palaeomagnetism to the study of continental displacements. Owing to the problems of resolving different components in these lavas it is suggested that more work should be done on baked laterites and on bauxites within the Northern Irish lava succession in order to improve presently available results. Some preliminary data from late Mesozoic and early Tertiary rocks of Europe (presently studied in our laboratory) seem to confirm the remagnetization hypothesis here laid down.

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